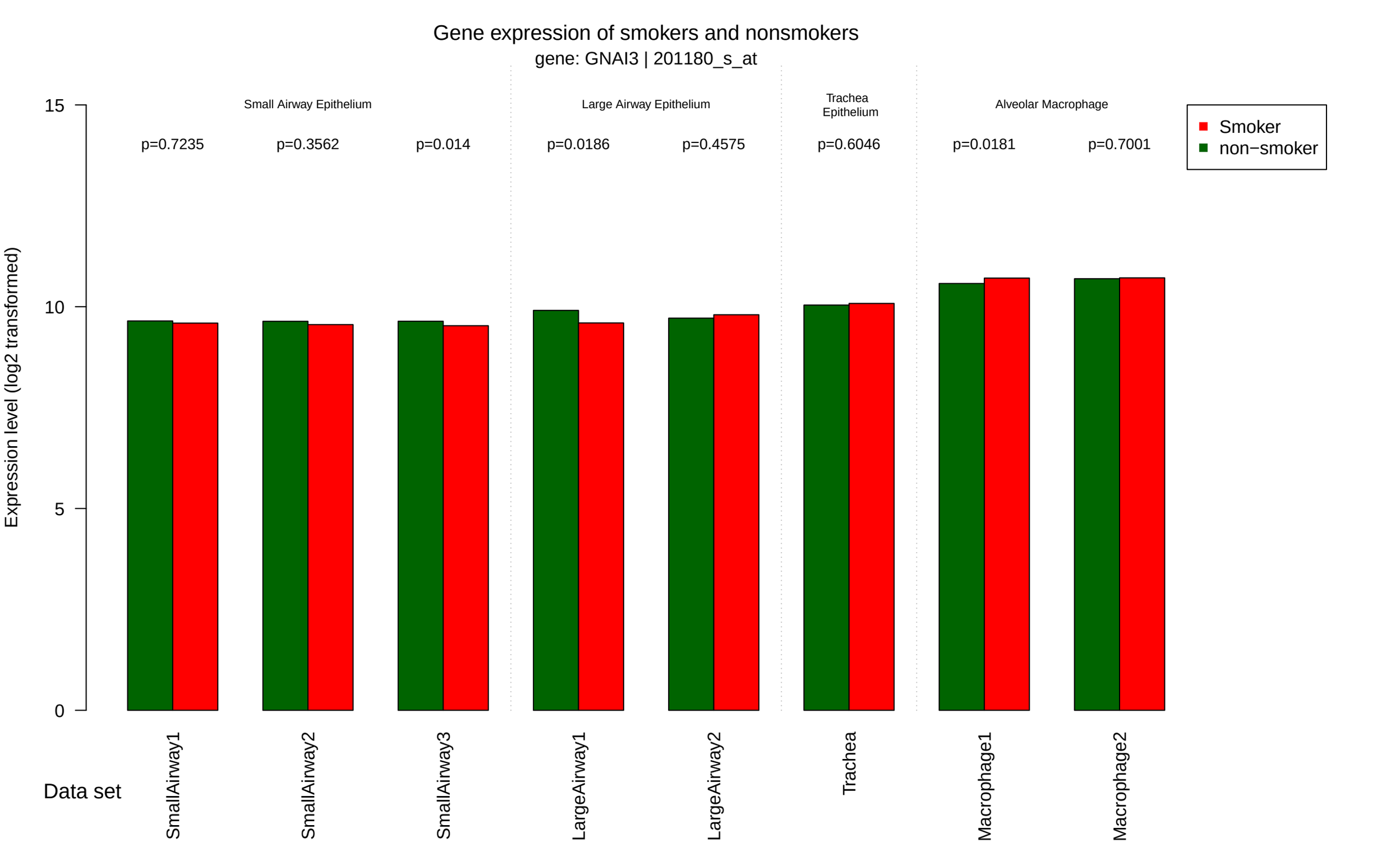
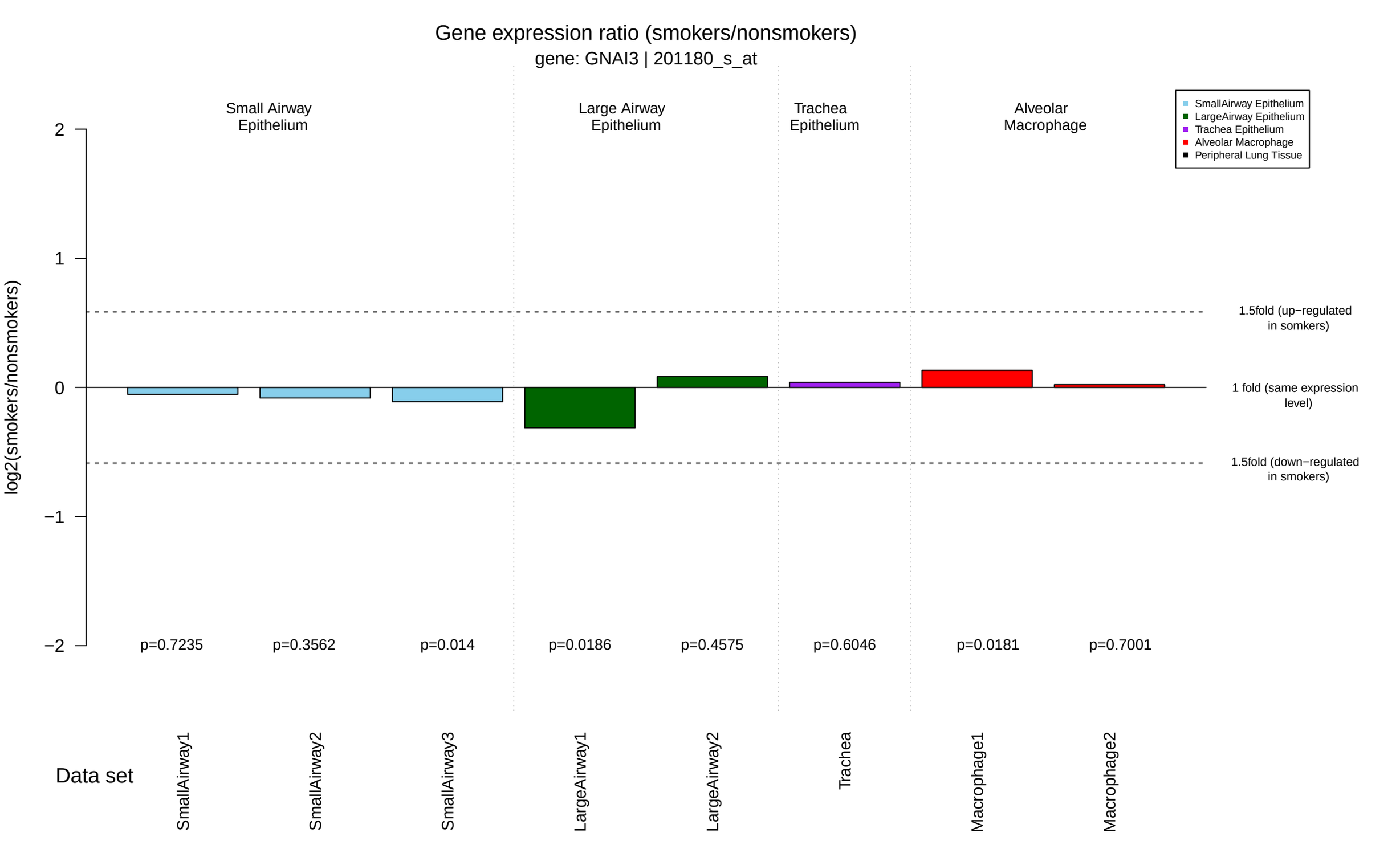
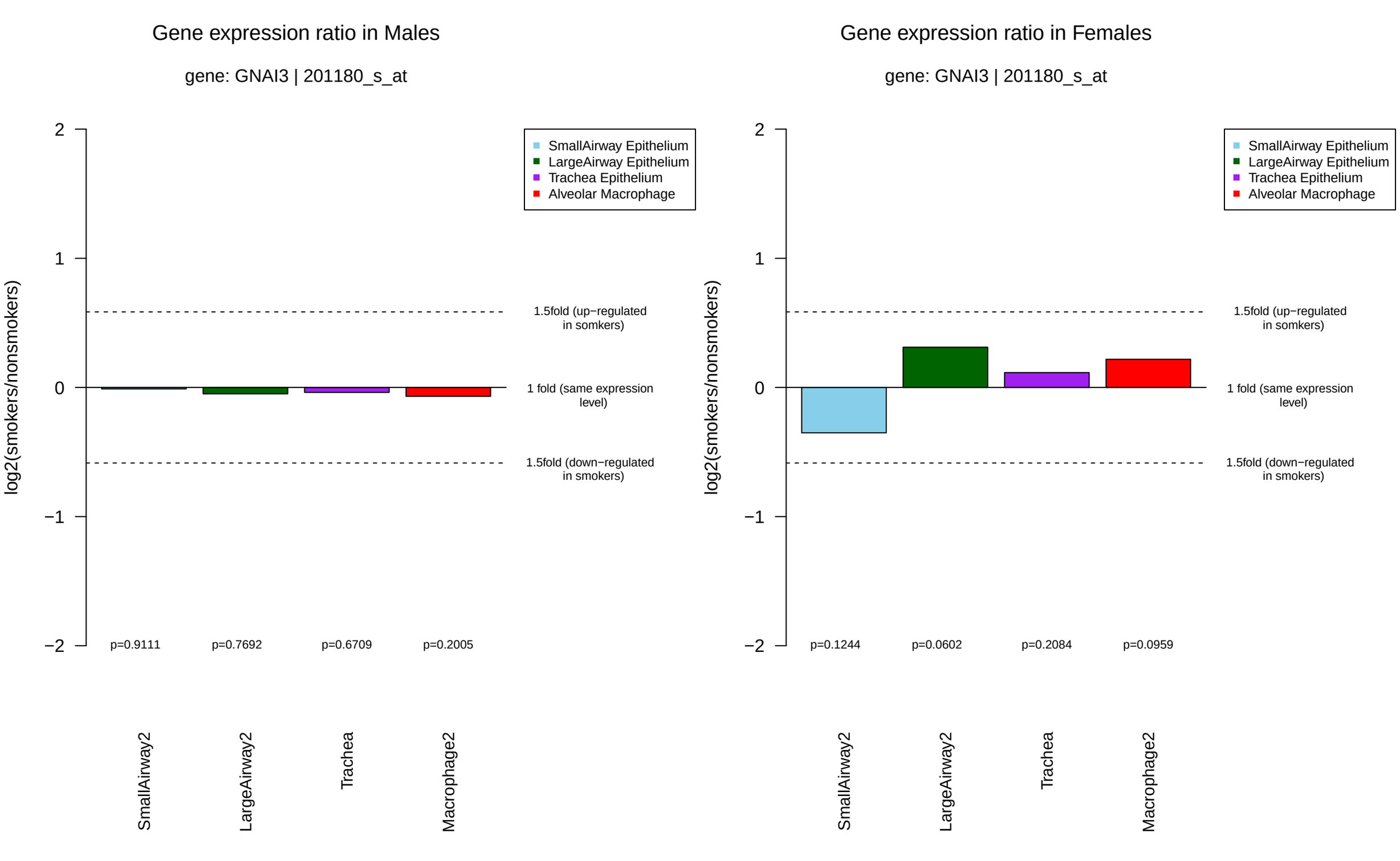
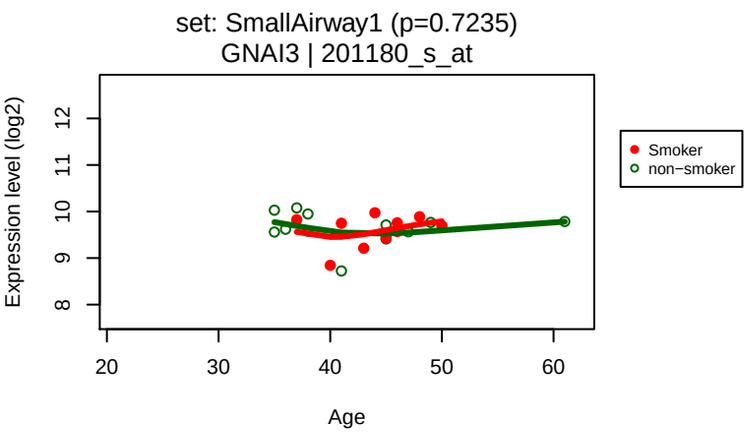
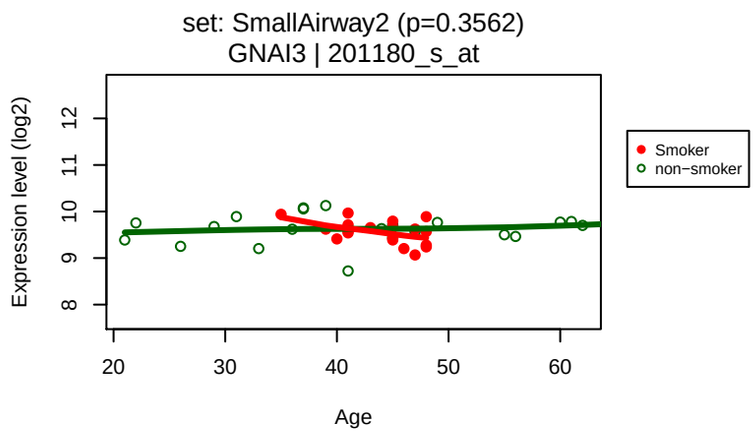
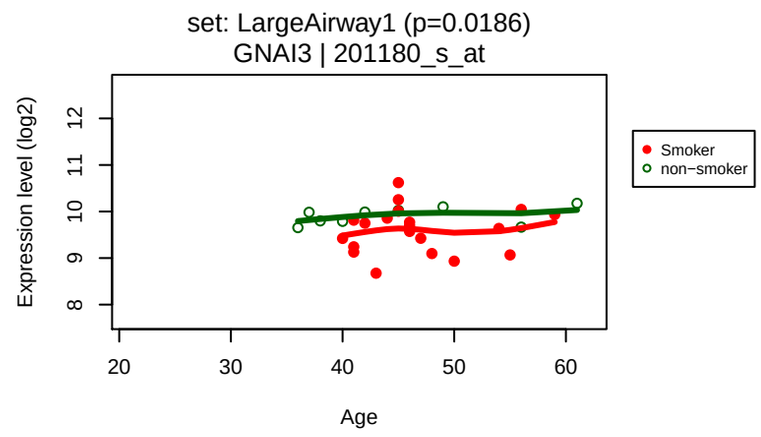

Gene expression of smokers and nonsmokers
gene: GNAI3 | 201180_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7235
p=0.3562
p=0.014
p=0.0186
p=0.4575
p=0.6046
p=0.0181
p=0.7001
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

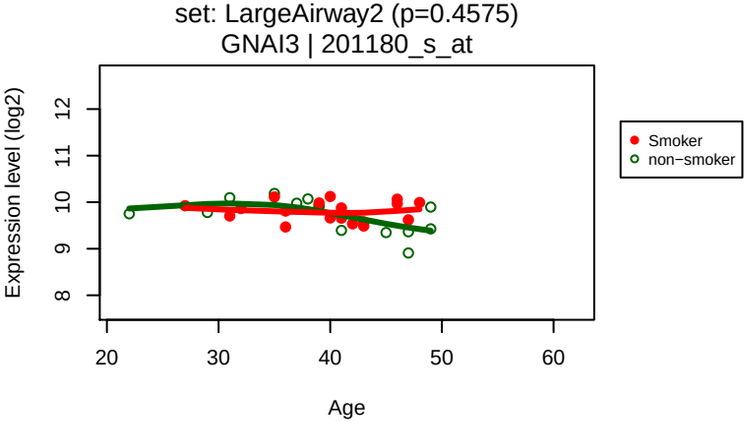

Gene expression ratio (smokers/nonsmokers)
gene: GNAI3 | 201180_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7235
p=0.3562
p=0.014
p=0.0186
p=0.4575
p=0.6046
p=0.0181
p=0.7001
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

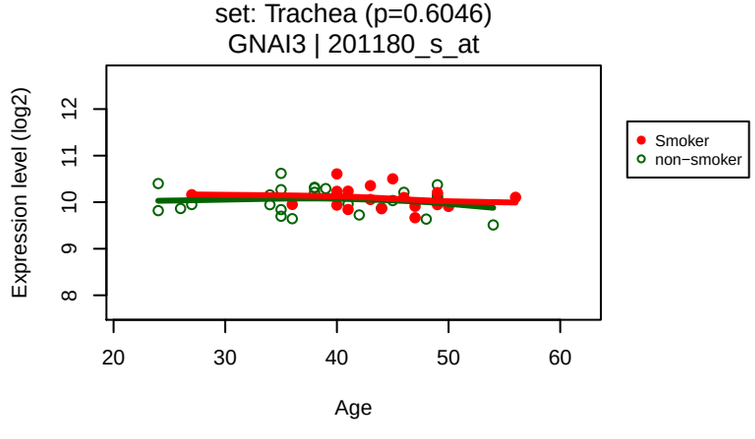

Gene expression ratio in Males
gene: GNAI3 | 201180_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9111
p=0.7692
p=0.6709
p=0.2005
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAI3 | 201180_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1244
p=0.0602
p=0.2084
p=0.0959
SmallAirway2
LargeAirway2
Trachea
Macrophage2

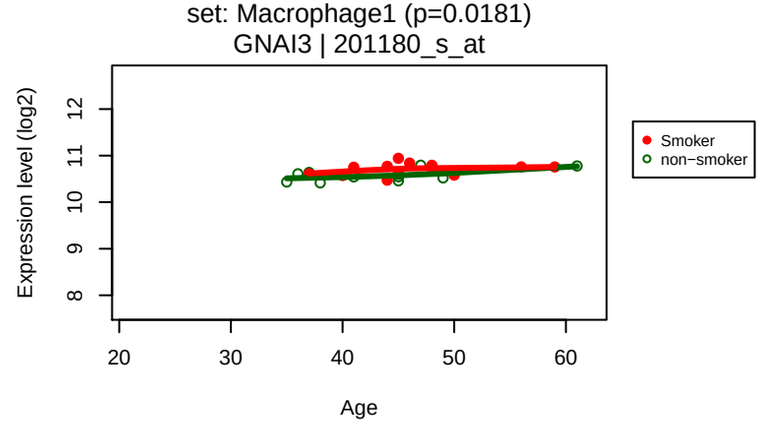

set: SmallAirway1 (p=0.7235)
GNAI3 | 201180_s_at
Expression level (log2)
Age
Smoker
non−smoker

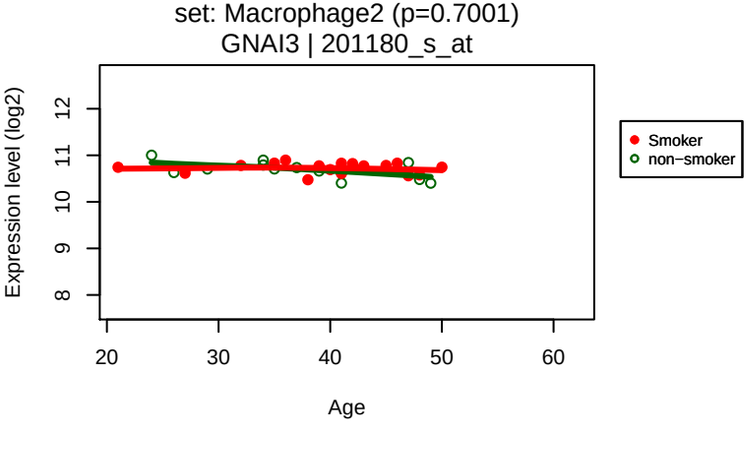

set: SmallAirway2 (p=0.3562)
GNAI3 | 201180_s_at
Expression level (log2)
Age
Smoker
non−smoker

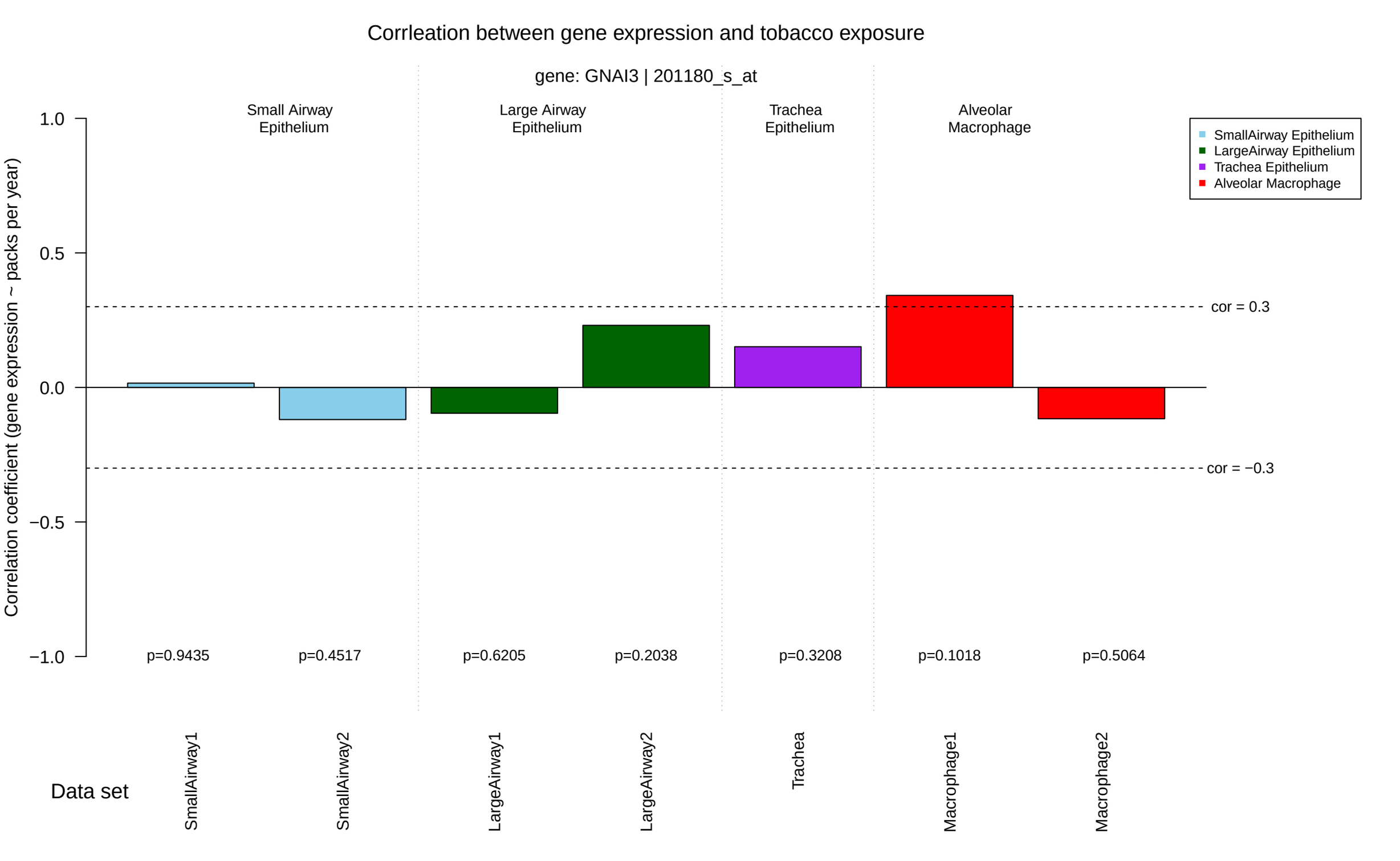

set: LargeAirway1 (p=0.0186)
GNAI3 | 201180_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4575)
GNAI3 | 201180_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6046)
GNAI3 | 201180_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0181)
GNAI3 | 201180_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7001)
GNAI3 | 201180_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAI3 | 201180_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9435
p=0.4517
p=0.6205
p=0.2038
p=0.3208
p=0.1018
p=0.5064
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2